

Ohm's Law and Series Circuits

Introductory physics / electronics · ~2 pages · for XR EduAgent study

Learning goals

You should be able to: (1) state Ohm's law; (2) relate voltage, current, and resistance; (3) compute equivalent resistance for two resistors in series; (4) predict how current changes when R increases.

Ohm's law

For many conductors at fixed temperature, $V = I R$, where V is voltage (volts), I is current (amperes), and R is resistance (ohms). Rearrangements: $I = V/R$ and $R = V/I$.

Intuition

Think of voltage as the “push,” resistance as how hard it is for charge to flow, and current as the flow rate. Higher R with the same V means smaller I.

Series resistors

In series, the same current flows through each resistor. Equivalent resistance is $R_{eq} = R_1 + R_2 (+ \dots)$. Total voltage divides across resistors proportional to their resistances.

A quick example

A 12 V battery, $R_1 = 2 \Omega$ and $R_2 = 4 \Omega$ in series $\rightarrow R_{eq} = 6 \Omega \rightarrow I = 12/6 = 2$ A. Voltage across R_1 is $I \cdot R_1 = 4$ V; across R_2 is 8 V.

Why interactive circuits help

Dragging resistance while watching current/voltage readouts reinforces $V = I R$ better than memorizing the equation alone.

Study use only · Short excerpt prepared for XR EduAgent usability study · Not a full textbook chapter.